A FILTERING FAMILY OF SUBFIELDS OF A POWER SERIES FRACTION FIELD WITH INTERSECTION THE P-TH POWERS

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- p-basis of a field of characteristic p. the statement counted by the sibling card S1-algebra-p-basis at 11362-11415. Supplies the indexing set and the description of k_J.
- Finiteness of a power series extension after a finite extension of the coefficient field. the author cites Algebra, Lemma algebra-lemma-finite-after-completion at proof lines 11537-11539. Gives finiteness of the ring extensions A_J to A.
- The intersection statement for filtering families of subfields, whose three hypotheses are verified here. the statement counted by the sibling card S1-algebra-intersection-subfields-subspace at 11417-11455, cited at proof lines 11542-11544. Supplies the list of properties to be checked and the use to which the family is put elsewhere.

Source reference. Original lemma 07P5: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 9785–9785. The lemma's own canonical tag, quoted so that readers can resolve the label inside the delivered text; it points at the counted statement itself and carries no proof content.

Source reference. Original lemma 07P4: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 9784–9784. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 0394: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 4259–4259. Navigation pointer to a statement the

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Lemma 0.2. *Let k be a field of characteristic $p > 0$. Let $\{x_i\}_{i \in I}$ be a p-basis for k. Let $n, m \geq 0$. Let K be the fraction field of $A = k[[x_1, \ldots, x_n]][y_1, \ldots, y_m]$. Let J be a finite subset of I. Consider the subfield $k/k_J/k^p$ generated by k^p and x_i with $i \in I \setminus J$. The fraction fields K_J of*

$$A_J = k_J[[x_1^p, \ldots, x_n^p]][y_1^p, \ldots, y_m^p]$$

form a family of subfields of K as in Lemma 07P4. Moreover, each of the ring extensions $A_J \subset A$ is finite.

Proof. Since k/k_J is finite, the ring extension $k_J[[x_1^p, \ldots, x_d^p]] \subset k[[x_1, \ldots, x_d]]$ is finite by Algebra, Lemma 0394. This implies that $A_J \to A$ is finite.

Let us check properties (1), (2), (3) of Lemma 07P4. Proof of (1). For $a \in A$ we see that $a^p \in A_J$. Hence $K^p \subset K_J$. Proof of (2). Suppose that $f/g^p \in K$, $f, g \in A$, $g \neq 0$ is contained in K_J for every choice of J. Fix J for the moment. Since $f/g^p \in K_J$ we can write $f/g^p = a/b^p$ with $a \in A_J$ and $b \in A$ nonzero. Hence $b^p f \in A_J$. For any A_J-derivation $D : A \to A$ we see that $0 = D(b^p f) = b^p D(f)$ hence $D(f) = 0$ as A is a domain. Taking $D = \partial_{x_i}$ and $D = \partial_{y_j}$ we conclude that $f \in k[[x_1^p, \ldots, x_n^p]][y_1^p, \ldots, y_d^p]$. Applying a k_J-derivation $\theta : k \to k$ we similarly conclude that all coefficients of f are in k_J, i.e., $f \in A_J$. Since it is clear that $A^p = \bigcap_J A_J$ where J ranges over all subfields as in the lemma we conclude $f \in A^p$ as desired. Proof of (3). This is clear because $K_{J \cup J'} \subset K_J \cap K_{J'}$. □